TARGET CASE STUDY

1.Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1.Data type of all columns in the "customers" table.

select column_name,data_type from `first-project-sql-458108.target.INFORMATION_SCHEMA.COLUMNS` where table_name='customers'

Row	column_name	data_type
1	customer_id	STRING
2	customer_unique_id	STRING
3	customer_zip_code_prefix	INT64
4	customer_city	STRING
5	customer_state	STRING

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2. Get the time range between which the orders were placed.

select min(order_purchase_timestamp) as order_placement_started_time,max(order_purchase_timestamp) as order_placement_stopped_time
from `target.orders`

order_placement_started_time	order_placement_stopped_time
2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC

INSIGHTS:
The dataset contains data for order placed between time period of 2016 september To 2018 october

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3. Count the Cities & States of customers who ordered during the given period.

select gl.geolocation_city as city,gl.geolocation_state as state,count(*) as count_of_locations from `target.customers` c join `target.orders` o
on c.customer_id=o.customer_id
join `target.geolocation` gl

```
on c.customer_zip_code_prefix=gl.geolocation_zip_code_prefix
where o.order_purchase_timestamp
between '2016-09-04 21:15:19' and '2018-10-17 17:30:18'
group by gl.geolocation_city,gl.geolocation_state
order by count_of_locations desc
```

ow	city	state	count_of_locations
1	rio de janeiro	RJ	1913894
2	sao paulo	SP	1164450
3	belo horizonte	MG	737556
4	niteroi	RJ	393175
5	curitiba	PR	255731
6	santos	SP	238952

INSIGHTS:

The highest number of orders were observed from the state of RJ, while the lowest number of orders came from state SP.

Within state SP, there is a wide variation — some cities recorded a high volume of orders, while others had very low order counts. This pattern was also seen in other states.

This indicates that the number of orders in a city is largely influenced by factors such as:

1.City size
2.Population
3.Income level and purchasing power of the residents

RECOMMENDATIONS:

We can provide special offers in smaller cities to help expand our product reach and attract a larger customer base.
While this may reduce income in the short term, as people experience and appreciate our service, they are likely to continue ordering even without offers in the future, leading to sustainable growth.

2.In-depth Exploration:

1.Is there a growing trend in the no. of orders placed over the past years?

```
select extract(year from order_purchase_timestamp) as year,
count(order_id) as number_of_orders
from `target.orders` group by year
order by year
```

w	year	number_of_orders
1	2016	329
2	2017	45101
3	2018	54011

INSIGHTS:

There is a clear **year-on-year growth in total orders**. The lower order volume in 2016 is due to limited data (only 8 months) and the platform being in its initial launch phase.

When analyzing quarterly trends:

1.**2016 and 2017** show a consistent **increase in orders over time** as the platform matured.

2.In **2018**, although the total number of orders is higher than in 2017, there is **no consistent growth across quarters** — in fact, a decline is observed in later quarters, indicating possible seasonality or a slowdown in momentum.

RECOMMENDATIONS:

1.Whenever a decline in order volume is observed, we should implement **targeted offers and promotions** to stimulate demand.

2.For example, during **Q2 of 2018**, timely promotional efforts could have helped mitigate the decline in orders in the following quarters.

3.Moving forward, this **proactive approach** should be consistently applied — by monitoring order trends closely and responding quickly with marketing and promotional strategies to maintain steady growth.

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2. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

```
select extract(month from order_purchase_timestamp) as month_no,
FORMAT_DATETIME('%B', order_purchase_timestamp) AS month_name,
extract(year from order_purchase_timestamp) as year,
count(order_id) as number_of_orders
from `target.orders` group by month_no,month_name,year
order by year,month_no
```

	month_no	month_name	year	number_of_orders
1	9	September	2016	4
2	10	October	2016	324
3	12	December	2016	1
4	1	January	2017	800
5	2	February	2017	1780
6	3	March	2017	2682

INSIGHTS:

Steady growth across months from Jan to August — in both 2017 and 2018 -> No strong negative seasonality seen in those months.

November 2017 peak — Possible seasonal spike (could be due to year-end sales, Black Friday, holiday season).

September 2018 and October 2018 → Unusual, sudden drop in orders

16 orders in Sept, 4 orders in Oct — inconsistent with previous trends.

This looks like a data issue or external factor — not seasonality (because Sept-Oct 2017 was normal).

RECOMMENDATIONS:

There is a noticeable drop in orders during September 2018. We should conduct a deeper analysis to understand the reasons behind this decline (such as seasonality, reduced marketing activity, operational issues, or external factors) in order to prevent similar drops in the future.

To address this, we can plan to:

1. Introduce targeted offers and promotions in September to boost demand.
2. Increase marketing efforts during this period to drive customer engagement.

3.Review any potential service or operational challenges that may have affected

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3.order volumes in that month.

During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

- 0-6 hrs : Dawn
- 7-12 hrs : Mornings
- 13-18 hrs : Afternoon
- 19-23 hrs : Night

```
select count(case when extract(hour from order_purchase_timestamp) between
0 and 6
then order_id end) as orders_in_Dawn,
count(case when extract(hour from order_purchase_timestamp) between
7 and 12
then order_id end) as orders_in_Morning,
count(case when extract(hour from order_purchase_timestamp) between
13 and 18
then order_id end) as orders_in_Afternoon,
count(case when extract(hour from order_purchase_timestamp) between
19 and 23
then order_id end) as orders_in_night
from `target.orders`
```

orders_in_Dawn	orders_in_Morning	orders_in_Afterno...	orders_in_night
5242	27733	38135	28331

INSIGHTS:

Order activity varies significantly by time of day. The highest number of orders are placed during the **afternoon**, followed by the **night**, then **morning**, with the lowest activity observed during **dawn**.

RECOMMENDATIONS:

People are actively placing orders during the **afternoon and night**. Running targeted ads during these peak hours can help **increase order volume** and **reach a larger audience** at the right time.

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3. Evolution of E-commerce orders in the Brazil region:

1. Get the month on month no. of orders placed in each state.

```
select extract(month from o.order_purchase_timestamp) as month_no,
FORMAT_DATETIME('%B', o.order_purchase_timestamp) AS month_name,
c.customer_state as state,
count(o.order_id) as count_of_orders
from `target.orders` o join `target.customers` c
on o.customer_id=c.customer_id
join `target.geolocation` gl
on c.customer_zip_code_prefix=gl.geolocation_zip_code_prefix
group by month_no,month_name,state
order by state,month_no
```

month_no	month_name	state	count_of_orders
1	January	AC	692
2	February	AC	512
3	March	AC	514
4	April	AC	785
5	May	AC	1154

<u>INSIGHTS:</u>

1. High-performing states

SP (São Paulo) has by far the highest number of orders every month — e.g. August peak: 660K orders.
RJ (Rio de Janeiro) and MG (Minas Gerais) are the next top states.
PR, RS, SC also contribute large consistent volumes.

2. Seasonality Trends

Many states see peak order volumes around May–August — possibly linked to mid-year sales, holiday shopping, or winter promotions in Brazil.
November also shows noticeable spikes (likely Black Friday sales).
September and October tend to show a drop in orders across most states — possible seasonal lull.

3. Underperforming states

RR, AC, AP, RO, PI, PB, SE, TO → have low order volumes throughout the year.
Opportunity to improve reach or target these regions more effectively.

4. Volatility in small states

Smaller states like RR, RO, AM show high variability month-to-month, which may indicate inconsistent marketing or operational challenges (e.g. logistics, delivery times).

5. Growth potential in mid-tier states

States like BA, CE, DF, GO, ES show strong volumes but have room for growth compared to SP and RJ.
Could benefit from regionalized campaigns.

RECOMMENDATIONS:

1.Capitalize on seasonality

Plan **major marketing campaigns** around May–August and November (Black Friday) where demand is already high.

Address slow months (Sep–Oct) with **special promotions** to smooth revenue.

2.Target underperforming regions

Run **localized marketing** in RR, AC, RO, SE, TO where orders are low.

Investigate potential barriers: **shipping cost, delivery times, awareness**.

3. Focus on repeat customers in top regions

SP, RJ, MG drive most of the business — invest in **loyalty programs, cross-sell/up-sell**, subscription models in these markets.

4. Strengthen operations/logistics

For highly volatile small states — improve **logistics partners**, optimize delivery times to build trust.

5. Regional campaign optimization

Tailor product recommendations by **state** — based on regional preferences.

Optimize **ad spend** using customer density (city + state level).

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2. How are the customers distributed across all the states?

```
select customer_state,count(customer_id) as number_of_customers
from `target.customers`
group by customer_state
order by number_of_customers desc
```

Job information | Results | Chart | JS

	customer_state	number_of_customers
1	SP	41746
2	RJ	12852
3	MG	11635
4	RS	5466
5	PR	5045

INSIGHTS:

state SP has the highest number of customers and state RR has the lowest number of customer.
When we analyse the difference between the two states we come to know that the SP is Brazil's richest and most industrialized state, huge population, very urban, economic hub of Brazil,
whereas RR is Least populated state, located in the Amazon region, economy based on agriculture, mining, and public sector

RECOMMENDATIONS:

1.For SP like state us peronalised digital marketing
Strengthen the brand present in this key market.

2.For state like RR offer affordable and essential products.
Build partnerships with local distributors.
Use community based marketing to grow awareness.

overall tailor strategies to local needs.
Apply learning from state SP to other regions where appropriate.

4. Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

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4. Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

1. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).
You can use the "payment_value" column in the payments table to get the cost of orders.

```
with year_wise_payment as
(
select extract(year from order_purchase_timestamp) as
year,round(sum(payment_value),2) as total_payment
from `target.orders` o join `target.payments` p
on o.order_id=p.order_id
where extract(year from order_purchase_timestamp) in (2017,2018) and extract(month
from order_purchase_timestamp) between 1 and 8
group by year
)
select max(case when year=2017 then total_payment end) as payment_2017,
max(case when year=2018 then total_payment end) as payment_2018,
round(((max(case when year=2018 then total_payment end) - max(case when
year=2017 then total_payment end))/max(case when year=2017 then total_payment
end))*100,2) as percent_increase
from year_wise_payment
```

payment_2017	payment_2018	percent_increase
3669022.12	8694733.84	136.98

<u>INSIGHTS:</u>

1. There is a **significant increase in total payment** from 2017 to 2018, with payment growth exceeding **100%**.

2.**January 2018** contributed the most to the overall increase in payments.

3.**August 2018** contributed the least to the year-over-year growth (compared to August 2017).

4.Within **2018 (Jan–Aug)**, the **month of February** recorded the **lowest payment amount** among those eight months.

<u>RECOMMENDATIONS:</u>

1. Leverage peak months (like January):

Strengthen marketing and promotional activities in January to maximize revenue during the highest-performing month.

2.Address low-performing months (February, August):

Investigate why **February 2018** had the lowest payment within the first 8 months — was it due to seasonality, customer behavior, or operational issues?

Consider targeted campaigns or discounts in **February and August** to boost sales and smooth out revenue fluctuations.

3.Monitor monthly trends:

Continuously track monthly payment trends to quickly identify and address unexpected dips or surges.

4.Resource planning:

Allocate operational resources (inventory, staffing, delivery capacity) in line with expected demand peaks and troughs.

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2. Calculate the Total & Average value of order price for each state.

```
select c.customer_state,round(sum(oi.price),2) as total_price,round(avg(oi.price),2) as average_price
from `target.geolocation` gl join `target.customers` c
on gl.geolocation_zip_code_prefix=c.customer_zip_code_prefix
join `target.orders` o
on c.customer_id = o.customer_id
join `target.order_items` oi
on o.order_id=oi.order_id
group by c.customer_state
order by total_price desc
```

ow	customer_state	total_price	average_price
1	SP	711841974.95	111.28
2	RJ	440143891.64	127.81
3	MG	397190155.87	121.18
4	RS	111180944.69	120.18
5	PR	85393530.14	119.21
6	SC	70665362.43	127.41

INSIGHTS:

1.The **total price** is highest for **state SP** and lowest for **state RR**.

2.The **average price per order** is lowest for **SP** and highest for **PB**.

3. SP having the highest total price but the lowest average price indicates that **SP has a very high number of orders**, compared to other states.

RECOMMENDATIONS:

Understand customer behavior in SP:

- Analyze why SP has a large volume of smaller-value orders — could be due to product mix, customer preferences, or purchasing patterns.
- Consider opportunities to **upsell or bundle** to increase average order value in SP.

Leverage high-value market in PB:

- PB shows high average price per order — explore strategies to further grow this premium segment (e.g., loyalty programs, targeted marketing).

Boost engagement in low-performing states:

- Develop campaigns to increase both order volume and value in **low total-price states** like RR.

Segment marketing and sales strategy by state:

- Tailor approaches — volume-driven strategy for SP, premium strategy for PB, growth strategy for RR.

---

3. Calculate the Total & Average value of order freight for each state.

```
select c.customer_state,round(sum(oi.freight_value),2) as
total_freight_value,round(avg(oi.freight_value),2) as average_freight_value
from `target.geolocation` gl join `target.customers` c
on gl.geolocation_zip_code_prefix=c.customer_zip_code_prefix
join `target.orders` o
on c.customer_id = o.customer_id
join `target.order_items` oi
on o.order_id=oi.order_id
group by c.customer_state
```

order by total_freight_value desc

Row	customer_state	total_freight_value	average_freight_value
1	SP	98575209.73	15.41
2	RJ	71967254.72	20.9
3	MG	67058347.09	20.46
4	RS	19910501.67	21.52
5	PR	14432377.78	20.15
6	SC	13472096.61	21.55

INSIGHTS:

1. The total freight value is highest for state SP and lowest for state RR.

2. The average freight value per order is lowest for SP and highest for PB.

3. SP having the highest total freight value but the lowest average freight value indicates that SP has many low-priced orders, or that customers are located close to stores, or that orders are easily transferable and may not require expensive packaging, compared to other states.

RECOMMENDATIONS:

1. **Optimize last-mile delivery in SP:**

Since SP likely benefits from short-distance or low-complexity deliveries, consider piloting **same-day or express delivery** services to gain a competitive edge.

2. Promote eco-friendly delivery options in SP:

Leverage SP's likely urban proximity to offer **bike or electric van deliveries**, reducing environmental impact and logistics costs further.

3. Adjust packaging strategies by region:

For states like SP with low average freight, test **lightweight or minimal packaging** options to cut costs and enhance sustainability.

In contrast, maintain **secure or premium packaging** in high-freight states like PB to match expectations.

4. Introduce tiered freight pricing or incentives:

Consider **freight-based incentives** (e.g., free shipping over a certain amount) to encourage higher order values, especially in SP where orders may be small.

5. Study freight patterns to improve route planning:

Use SP as a **model state** to analyze how urban density and proximity affect freight efficiency, and apply learnings to optimize routes in similar urban regions.

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5. Analysis based on sales, freight and delivery time.

1. Find the no. of days taken to deliver each order from the order's purchase date as delivery time.
Also, calculate the difference (in days) between the estimated & actual delivery date of an order.
Do this in a single query.

You can calculate the delivery time and the difference between the estimated & actual delivery date using the given formula:

- time_to_deliver = order_delivered_customer_date - order_purchase_timestamp
- diff_estimated_delivery = order_delivered_customer_date - order_estimated_delivery_date

```
select p.`product category`,o.order_id,DATE_DIFF(order_delivered_customer_date,
order_purchase_timestamp, DAY) as no_of_days_to_deliver,
DATE_DIFF(order_delivered_customer_date, order_estimated_delivery_date, DAY) as
no_of_days_diff_estimated_delivery
from `target.orders` o join `target.order_items` oi
on o.order_id= oi.order_id
join `target.products` p
on oi.product_id=p.product_id
where o.order_status='delivered'
order by no_of_days_to_deliver desc
```

Row	order_id	no_of_days_to_deliver	no_of_days_diff_estimated_delivery
1	ca07593549f1816d26a572e06...	209	181
2	1b3190b2dfa9d789e1f14c05b6...	208	188
3	440d0d17af552815d15a9e41a...	195	165
4	285ab9426d6982034523a855f...	194	166
5	0f4519c5f1c541ddec9f21b3bd...	194	161
6	2fb597c2f772eca01b1f5c561bf...	194	155

<u>INSIGHTS:</u>

1. There are some missing entries for order_delivered_customer_date, indicating uncertainty about whether a customer has received the product. However, this is relatively low — less than 1%.

2. In 3% of cases, the estimated delivery date exactly matched the actual delivery date.

3. In 7% of cases, orders were delivered later than the estimated delivery date.

4.In 91% of cases, orders were delivered earlier than the estimated delivery date.

5. No consistent patterns were found in the 7% of late deliveries when analyzed by product category or region.

RECOMMENDATIONS:

1. Investigate missing delivery date entries to ensure accurate delivery tracking and improve data quality.

2. Since 91% of orders are being delivered earlier than estimated, review and adjust delivery date estimation models to better reflect actual delivery times — this can help set more accurate customer expectations.

3. Communicate consistently with customers about delivery performance — highlighting the high rate of early deliveries could help improve customer satisfaction and perception of service reliability.

---

2. Find out the top 5 states with the highest & lowest average freight value.

```
with avg_freight_tab as
(
select c.customer_state,round(avg(oi.freight_value),2) as avg_freight_value
from `target.customers` c join `target.orders` o
on c.customer_id=o.customer_id
join `target.order_items` oi
on oi.order_id=o.order_id
group by c.customer_state
order by avg_freight_value
)
(select *,'top_5_highest_freight_value' as category from avg_freight_tab order by
avg_freight_value desc limit 5)
union all
(select *,'top_5_lowest_freight_value' as category from avg_freight_tab order by
avg_freight_value asc limit 5)
```

customer_state	avg_freight_value	category
RR	42.98	top_5_highest_freight_value
PB	42.72	top_5_highest_freight_value
RO	41.07	top_5_highest_freight_value
AC	40.07	top_5_highest_freight_value
PI	39.15	top_5_highest_freight_value
SP	15.15	top_5_lowest_freight_value

INSIGHTS:

The **top 5 states with the highest average freight value** are: **RR, PB, RO, AC, PI**.
The **top 5 states with the lowest average freight value** are: **SP, PR, MG, RJ, DF**.

Common characteristics of high average freight value states (RR, PB, RO, AC, PI):

1.Smaller markets

2.Lower urbanization

3.Low consumer spending

4.Low order volume

5.Higher freight cost (remote locations)

Common characteristics of low average freight value states (SP, PR, MG, RJ, DF):

1. Large urban populations

2. Developed infrastructure

3. High consumer spending

4. High order volume

5. More efficient logistics

RECOMMENDATIONS:

1. **Optimize shipping strategies** for high-freight states by exploring local fulfillment centers, bulk shipping, or partnerships with regional carriers to reduce costs.

2. **Analyze product mix and packaging** for high-freight states to identify opportunities for lighter, more compact packaging to lower shipping costs.

3. **Prioritize marketing and promotions** in low-freight states (SP, PR, MG, RJ, DF) to leverage their cost-efficient logistics and high consumer demand.

4. **Conduct targeted logistics improvements** in underperforming high-freight states to improve delivery speed and reduce customer churn.

5. **Monitor freight costs over time** to ensure the freight strategy remains competitive as regional infrastructure and consumer trends evolve

---

3. Find out the top 5 states with the highest & lowest average delivery time.

```
with delivery_duration as
(
select c.customer_state,
round(AVG(TIMESTAMP_DIFF(o.order_delivered_customer_date,o.order_purchase_time
stamp, HOUR)),2) as
avg_delivery_time_in_hours
from `target.customers` c join `target.orders` o
on c.customer_id = o.customer_id
group by c.customer_state
)
(select *,'top_5_highest_delivery_duration' as category from delivery_duration order by
avg_delivery_time_in_hours desc limit 5)
union all
(select *,'top_5_lowest_delivery_duration' as category from delivery_duration order by
avg_delivery_time_in_hours asc limit 5)
```

customer_state	avg_delivery_time...	category
RR	704.73	top_5_highest_delivery_duration
AP	651.97	top_5_highest_delivery_duration
AM	633.7	top_5_highest_delivery_duration
AL	588.54	top_5_highest_delivery_duration
PA	570.05	top_5_highest_delivery_duration
SP	209.77	top_5_lowest_delivery_duration

INSIGHTS:

The average delivery time is highest in the states: RR, AP, AM, AL, PA.

These are generally remote states with challenging geography or lower infrastructure levels, which may contribute to longer delivery times.

The average delivery time is lowest in the states: SP, PR, MG, DF, SC.

These are more urbanized states with better transportation networks and higher logistics efficiency, resulting in faster deliveries.

RECOMMENDATIONS:

1. **Investigate logistics bottlenecks** in high delivery time states (RR, AP, AM, AL, PA), such as transport availability, carrier options, or last-mile challenges.

2. **Explore partnerships with regional carriers** or use local fulfillment centers to improve delivery times in remote areas.

3. **Promote fast delivery options** in states with low average delivery times (SP, PR, MG, DF, SC) to attract more customers.

4. **Monitor customer satisfaction** in high delivery time states and offer incentives (such as free shipping or discounts) to offset longer wait times.

5. **Use this insight to inform delivery time expectations** on your e-commerce site, providing accurate estimates by region.

---

4. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.
You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

```
SELECT
c.customer_state,round(AVG(TIMESTAMP_DIFF(o.order_delivered_customer_date,
o.order_estimated_delivery_date, DAY)),0) AS avg_diff_days
FROM `target.orders` o JOIN `target.customers` c
ON o.customer_id = c.customer_id
GROUP BY
c.customer_state
order by avg_diff_days asc
limit 5
```

	customer_state	avg_diff_days
1	AC	-20.0
2	RO	-19.0
3	AP	-19.0
4	AM	-19.0
5	RR	-16.0

INSIGHTS:

In the states **AC, RO, AP, AM, and RR**, orders are consistently delivered **earlier than the estimated delivery date** (negative average delivery delay).
This suggests that either:

1. The estimated delivery windows in these regions are **too conservative**, or

2. The local delivery partners and logistics are performing **better than expected**.

Possible contributing factors:

1. Lower order volumes → faster handling

2. Shorter delivery routes in these states

3. Less traffic congestion compared to major urban centers

4. More efficient local courier services

RECOMMENDATIONS:

1. Optimize Estimated Delivery Times
Review and adjust the estimated delivery date logic for these states to better match actual delivery patterns.
This will improve **customer satisfaction** by providing more accurate expectations.

2. Promote Early Delivery as a Feature
Highlight "faster than promised" delivery performance in customer communications — this can enhance trust and encourage repeat orders.

3. Analyze Operational Best Practices
Study what's working well in these regions — is it the carrier? the route planning?
Apply these learnings to underperforming states to improve overall delivery speed.

4. Cost Efficiency Opportunities
If consistent early delivery leads to higher costs (e.g. premium delivery partners), evaluate whether slight adjustments to estimated delivery windows can **optimize costs** without hurting customer experience.

---

6. Analysis based on the payments

1. Find the month on month no. of orders placed using different payment types.

```
select extract(month from  order_purchase_timestamp) as month,round(sum(case
when payment_type='credit_card' then payment_value end),2) as
total_credit_card_payment,
round(sum(case when payment_type='UPI' then payment_value end),2) as
total_UPI_payment,
round(sum(case when payment_type='voucher' then payment_value end),2) as
total_voucher_payment,
round(sum(case when payment_type='debit_card' then payment_value end),2) as
total_debit_card_payment,
round(sum(case when payment_type='not_defined' then payment_value end),2) as
total_payment_not_defined
from `target.orders` o join `target.payments` p
on o.order_id=p.order_id
group by month order by month
```

formation | Results | Chart | JSON | Execution details | Execution graph

month	total_credit_card...	total_UPI_payment	total_voucher_pa...	total_debit_card_...	total_payment_not_defined
1	978496.06	228919.09	33789.99	12287.08	null
2	1005556.56	240589.46	29245.48	8979.85	null
3	1288259.04	273698.55	35590.22	11967.91	null
4	1256393.72	274854.33	33752.93	13572.53	null
5	1372603.66	324031.83	37531.33	12734.15	null
6	1199776.83	261124.44	36158.82	38096.79	null

INSIGHTS:

1. Across all months, **Credit Card** remains the most dominant payment method, contributing the highest share of payments consistently.

2. **UPI** is the second most used method and shows steady usage across months — indicating growing customer preference for digital wallets and quick payments.

3. **Voucher** and **Debit Card** usage ranks lower, and their order fluctuates month to month — possibly influenced by promotional campaigns or seasonal offers.

4. Other payment methods (if any) contribute minimally to the overall share.

RECOMMENDATIONS:

1. Encourage UPI Adoption:
Since UPI is already growing and popular, consider **incentivizing UPI payments** with

small cashback offers to further shift users toward low-cost, real-time payment channels.

2. Credit Card Rewards:
Maintain and promote partnerships that provide **rewards on credit card transactions** (e.g. points, discounts), to retain and deepen engagement with this high-value customer segment.

3. Analyze Voucher Trends:
Investigate the specific campaigns or triggers that drive Voucher usage peaks — optimize timing and targeting of such offers to improve their ROI.

4. Simplify Debit Card Experience:
Explore if users face any friction when paying by debit card — **streamlining the process** may increase its usage and give customers more choice.

5. Monitor Fraud Risk:
Since Credit Card payments dominate, ensure that **fraud detection and prevention measures** for card transactions remain strong to avoid potential risk concentration.

6. Payment Method Personalization:
On checkout pages, consider **personalizing the order of payment options** based on user behavior — improving convenience and conversion rates.

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2. Find the no. of orders placed on the basis of the payment installments that have been paid.

```
select p.payment_installments,count(o.order_id) as number_of_orders
from `target.orders` o join `target.payments` p
on o.order_id=p.order_id
group by p.payment_installments
order by p.payment_installments
```

payment_installments	number_of_orders
0	2
1	52546
2	12413
3	10461
4	7098
5	5239

<u>INSIGHTS:</u>

1. Majority of orders (78%) are paid with **short-term installments** of **1 to 4 installments**:

1 installment: **52,546 orders** (the largest share by far)

2 installments: 12,413 orders

3 installments: 10,461 orders

4 installments: 7,098 orders

2. Single payment dominates:
More than half of all customers prefer to **pay in full (1 installment)** — showing strong trust in the platform and customer willingness to avoid debt.

3. Significant spike at 10 installments (5,328 orders) — showing a clear customer segment that prefers **longer-term financing** for higher-value purchases (likely big-ticket items).

4. Installments between 5–9 are used but not dominant, suggesting:

Either these options are less promoted

Or customers clearly prefer either **short-term (1–4)** or **long-term (10)** options.

5. Very few customers use **very long installment plans** (12, 15, 18, 24 months), showing limited demand for ultra-long financing.

RECOMMENDATIONS:

1. Highlight "1-Click Full Payment"
Since **most customers prefer 1 installment**, emphasize this as the fastest checkout option to further reduce cart abandonment.

2. Promote 2–4 Installment Offers
These options are also popular — consider offering **0% EMI** for 2–4 months to further boost order volume and customer satisfaction.

3. Leverage 10-installment demand
For higher-value products (electronics, furniture), actively promote **10-installment plans** — this is a sweet spot for long-term financing based on current customer behavior.

4.Re-evaluate 5–9 Installment Plans
Usage of 5–9 installments is much lower. Consider:

Reviewing if interest rates are too high

Simplifying these options (perhaps just offering 6 or 9-month plans)

Making marketing clearer for mid-term EMI options.

5.Simplify Options on Checkout
Too many installment options (up to 24) could confuse customers. You can **prioritize** and display:

1, 2, 3, 4, 6, 10 installments more prominently

Hide or move the rarely used options (13–24) under an "advanced" link.

6.Segment Campaigns:
Customers choosing **10-installments** may respond well to targeted **high-ticket offers or financing promotions**.
Customers using **1–2 installments** may be more price-sensitive — target them with discounts or loyalty points.